



IT SIMPLERA INSTITUTE INTERNSHIP PROGRAM

WEEK 01 REPORT

Wazuh SIEM Deployment and Configuration

Submitted By

Name: Challa Madhumitha

Reg No: SOCB01-2121

Role: Soc Analyst Intern

Submission Date: July 2026

Instructor: Zia Ullah

TABLE OF CONTENTS

S.No	Contents	Page No.
1	Introduction	3
2	Objective	3
3	Software and System Requirements	3
4	Wazuh SIEM Deployment	4
5	Windows Agent Installation	5
6	Linux Agent Installation	6
7	File Integrity Monitoring (FIM)	8
8	Security Event Generation	8
9	Alert Verification	9
10	Results	10
11	Challenges Faced	11
12	Learning Outcomes	11
13	Conclusion	12

1. Introduction

In today's digital environment, organizations generate a massive amount of security-related data from servers, workstations, applications, and network devices. Monitoring these logs manually is inefficient and increases the risk of missing critical security incidents. Security Information and Event Management (SIEM) solutions address this challenge by collecting, analyzing, and correlating logs from multiple systems in a centralized platform.

Wazuh is an open-source SIEM and Extended Detection and Response (XDR) platform widely used by Security Operations Centers (SOCs). It provides capabilities such as log collection, endpoint monitoring, vulnerability detection, malware detection, configuration assessment, and File Integrity Monitoring (FIM). These features help security teams detect suspicious activities, investigate incidents, and respond effectively.

The objective of this practical task was to deploy the Wazuh Virtual Appliance, configure the Wazuh Manager, install Wazuh Agents on Windows and Linux systems, enable File Integrity Monitoring, generate security events, and validate that alerts were successfully displayed in the Wazuh Dashboard. This exercise provided valuable hands-on experience in building and operating a basic SOC monitoring environment.

2. Objective

The primary objectives of this project were to:

- Deploy the Wazuh Virtual Appliance in a virtualized environment.
- Configure the Wazuh Manager for centralized monitoring.
- Install and configure the Wazuh Agent on both Windows and Linux endpoints.
- Register the agents with the Wazuh Manager.
- Verify successful communication between endpoints and the manager.
- Configure File Integrity Monitoring (FIM) for selected directories.
- Generate security events through file modifications.
- Observe and analyze alerts generated in the Wazuh Dashboard.
- Gain practical knowledge of endpoint monitoring and SOC operations.
- Document the complete implementation process with screenshots.

3. Software and System Requirements

The following software and resources were used to complete this project:

Hardware Requirements

- Laptop/Desktop Computer
- Minimum 8 GB RAM
- Stable Internet Connection

Software Requirements

- Oracle VirtualBox / VMware Workstation
- Wazuh Virtual Appliance
- Windows Operating System
- Linux Operating System (Ubuntu/Kali)
- Modern Web Browser
- Wazuh Agent Packages

The virtualization software was used to host the Wazuh appliance, while Windows and Linux systems acted as monitored endpoints connected to the Wazuh Manager.

4. Wazuh SIEM Deployment

The first step of the implementation was deploying the Wazuh Virtual Appliance. The appliance image was imported into the virtualization software and configured with the recommended hardware resources. Once the virtual machine booted successfully, the assigned IP address was identified using the system console.

The Wazuh Dashboard was then accessed through a web browser using the appliance IP address. After logging into the dashboard, the main interface displayed the health status of the Wazuh Manager and provided access to monitoring modules such as Security Events, Agents, Vulnerability Detection, File Integrity Monitoring, Threat Intelligence, and Dashboard Analytics.

The successful deployment confirmed that the SIEM platform was ready to receive logs and monitor connected endpoints.

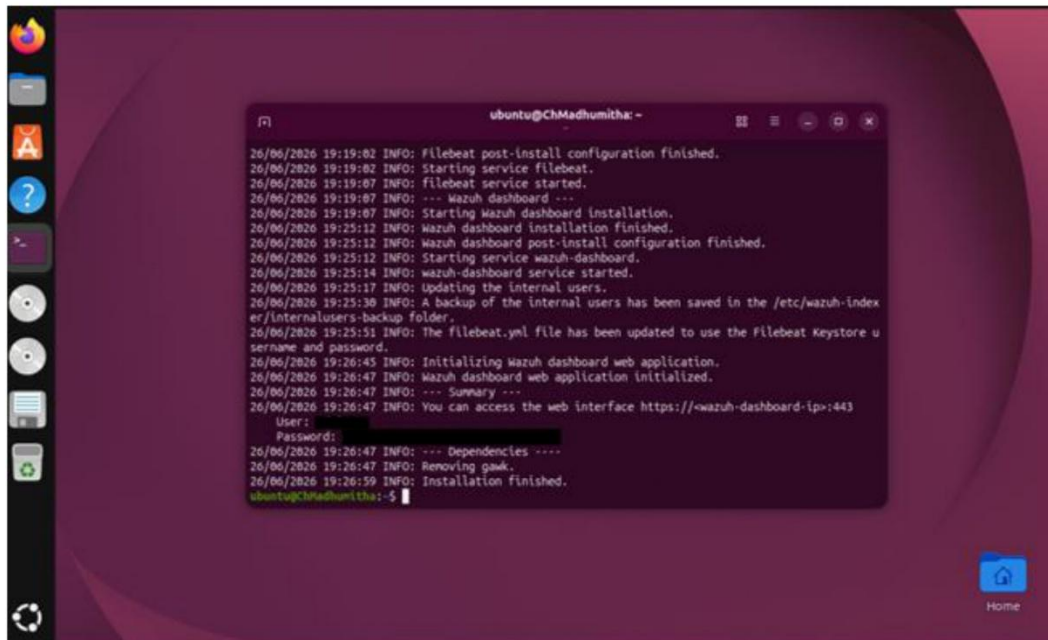


Figure 1 Wazuh Virtual Appliance Running

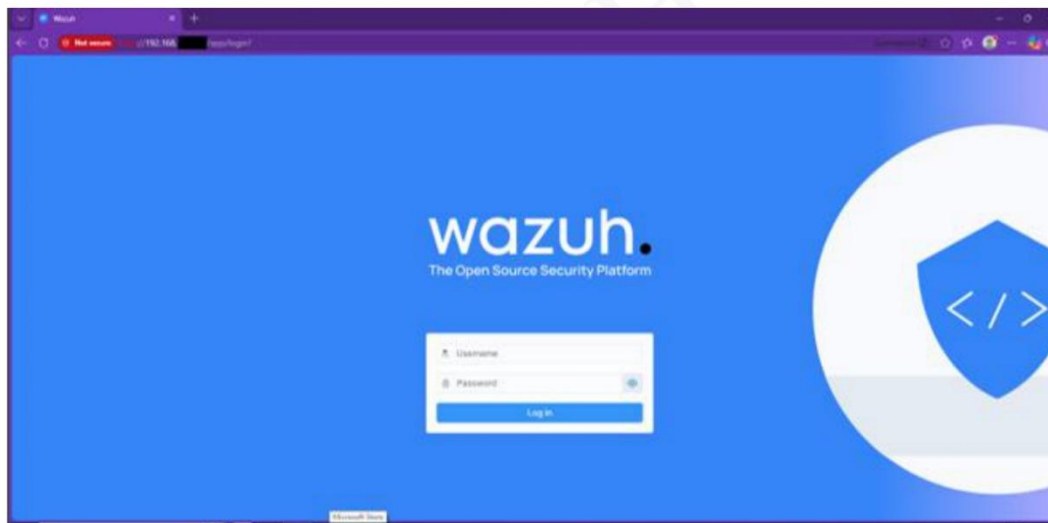


Figure 2 Wazuh Dashboard Login

5. Windows Agent Installation

The Wazuh Agent package for Windows was downloaded and installed on the Windows endpoint. During the installation process, the IP address of the Wazuh Manager and the Agent Name were specified to establish communication with the centralized monitoring server.

After the installation completed successfully, the Wazuh Agent service was started. The endpoint automatically registered with the Wazuh Manager and appeared under the Agents section of the dashboard.

The registration status changed to Active, confirming that the Windows endpoint was successfully connected and capable of sending logs and security events to the manager.

This configuration enables continuous monitoring of Windows security events, system activity, and file changes.

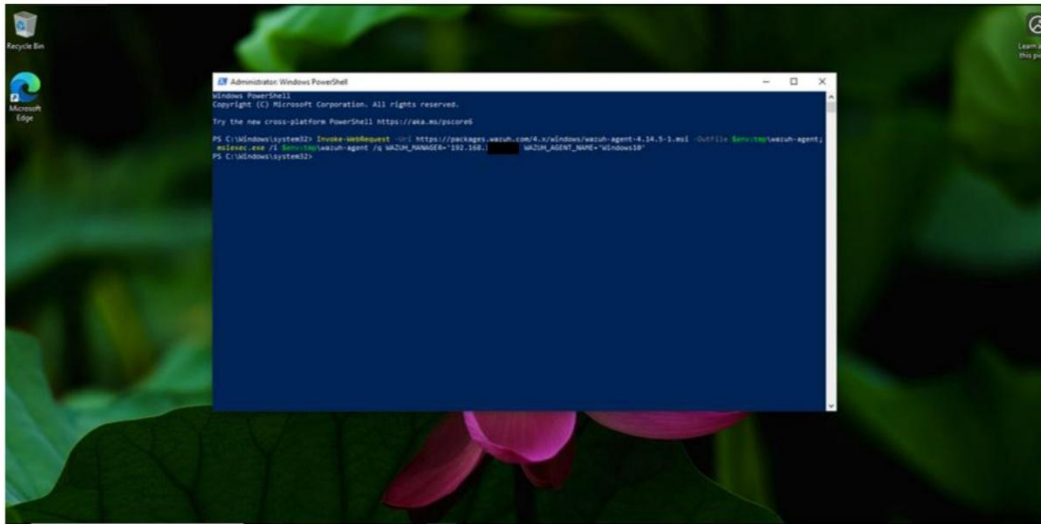


Figure 3 Windows Agent Installation

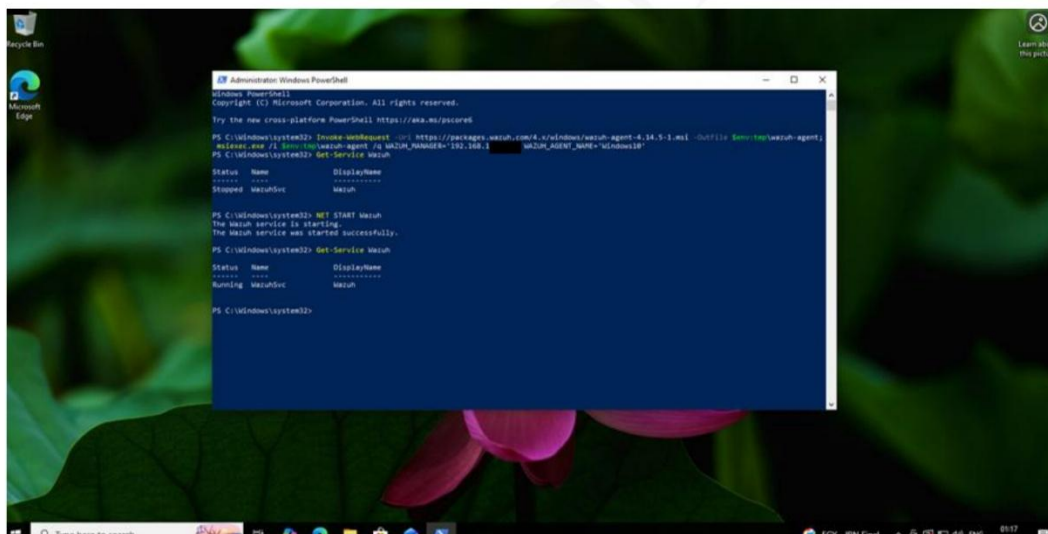


Figure 4 Windows Agent Registered Successfully

6. Linux Agent Installation

The Wazuh Agent was installed on the Linux system using the terminal. After installing the required package, the agent configuration file was updated with the IP address of the Wazuh Manager.

The Wazuh Agent service was enabled to start automatically during system boot and then started manually. Once the service became active, the Linux endpoint successfully registered

with the Wazuh Manager. The dashboard displayed the Linux endpoint as an active agent, indicating successful communication between the manager and the monitored system.

Monitoring Linux systems is essential for detecting unauthorized file changes, privilege escalation attempts, configuration modifications, and suspicious system activities.

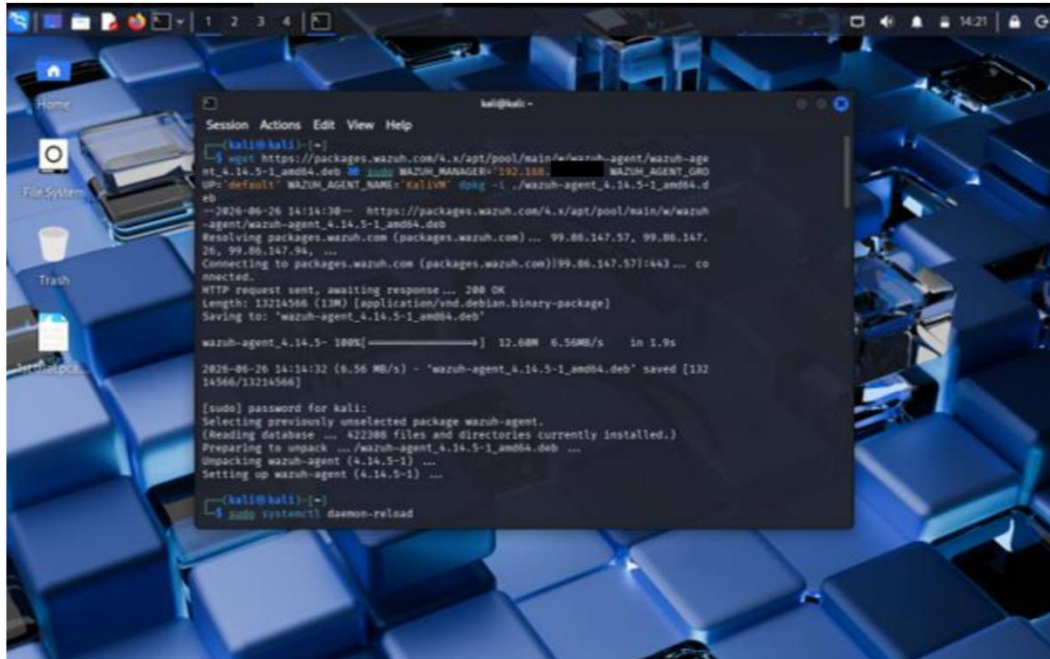


Figure 5 Linux Agent Installation

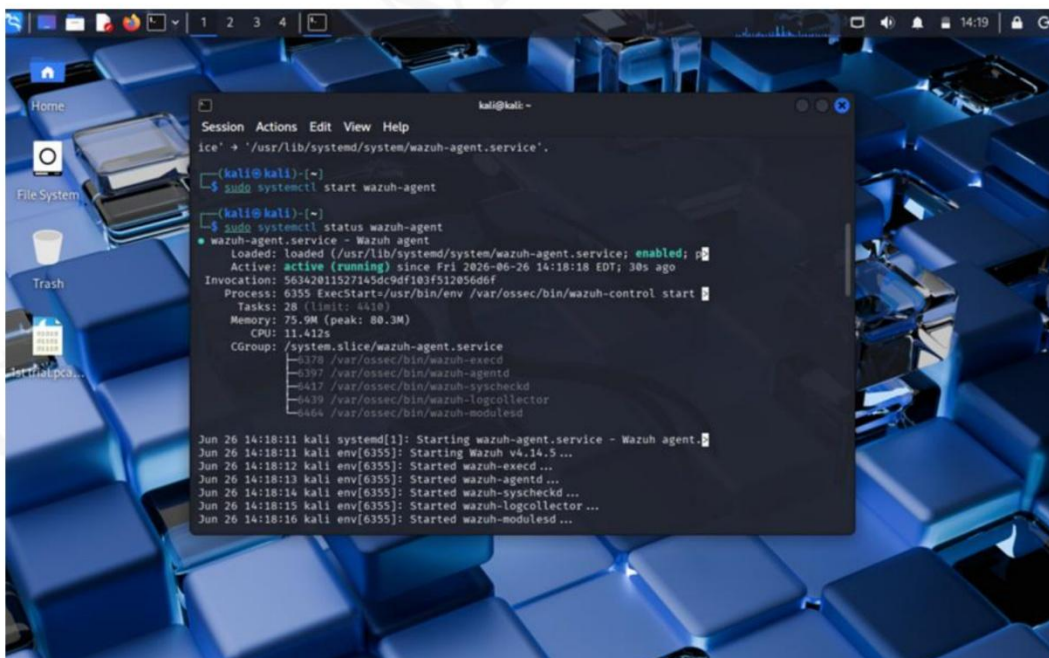


Figure 6 Linux Agent Connected to Wazuh

7. File Integrity Monitoring (FIM)

File Integrity Monitoring (FIM) is an important security capability that continuously monitors selected files and directories for unauthorized modifications. It helps detect attacks such as ransomware activity, unauthorized configuration changes, malware infections, and insider threats.

For this project, File Integrity Monitoring was configured by defining important directories in the Wazuh Agent configuration file. Typical monitored locations included user directories, system configuration folders, and other sensitive locations.

After updating the configuration, the Wazuh Agent service was restarted to apply the changes. The configuration was then validated through the Wazuh Dashboard to ensure that monitored directories were actively being tracked.

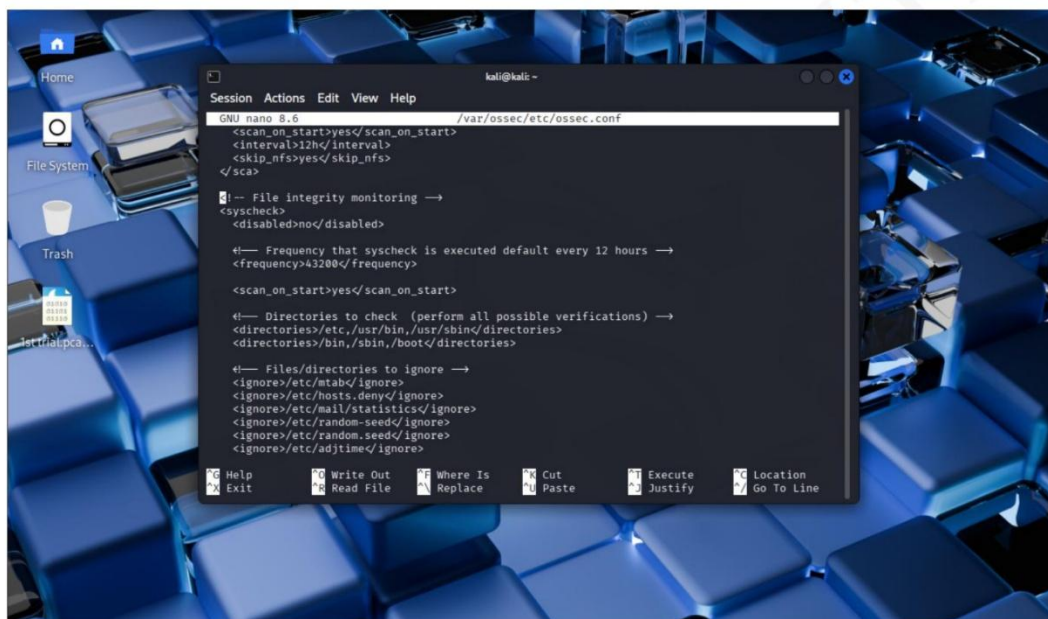


Figure 7 FIM Configuration

8. Security Event Generation

To verify the effectiveness of File Integrity Monitoring, several controlled file operations were performed on the monitored directories.

The following activities were executed:

- Created a new text file.
- Modified the file contents.
- Renamed the file.
- Deleted the file.

Each of these actions generated security events that were collected by the Wazuh Agent and forwarded to the Wazuh Manager for analysis.

This practical testing demonstrated that Wazuh continuously monitors file activities and reports any unauthorized or unexpected modifications in real time.

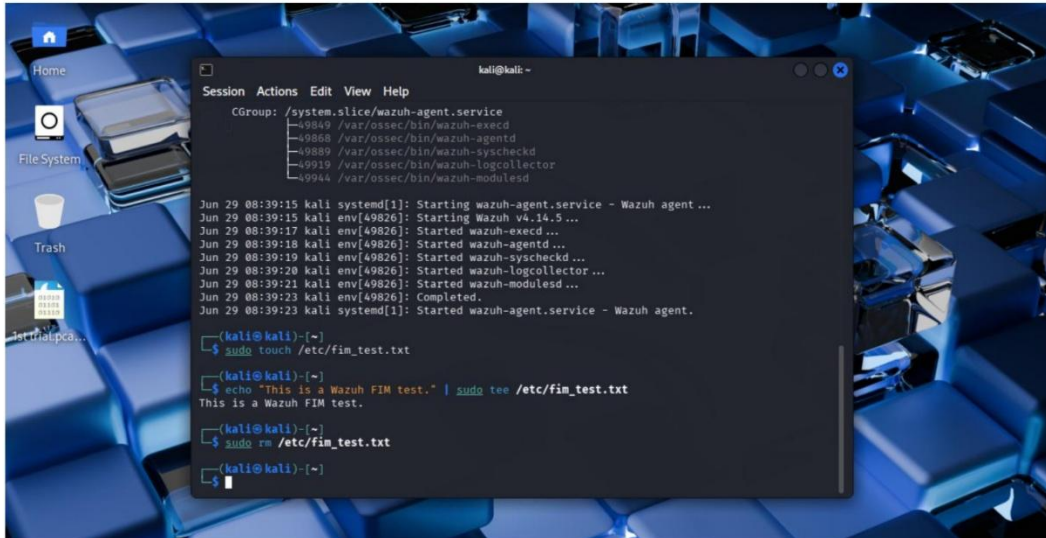


Figure 8 Test File Operations

9. Alert Verification

Following the file operations, the generated events were successfully displayed in the Wazuh Dashboard under the Security Events section.

Each alert contained valuable security information including:

- Agent Name
- Event Timestamp
- Rule ID
- Rule Description
- Alert Severity
- File Path
- Event Type (Created, Modified, Deleted)

The successful generation of alerts confirmed that the Wazuh Manager correctly processed the events received from both Windows and Linux endpoints.

This step validated the complete monitoring workflow from endpoint activity to centralized alert generation, which is a core responsibility of a Security Operations Center.

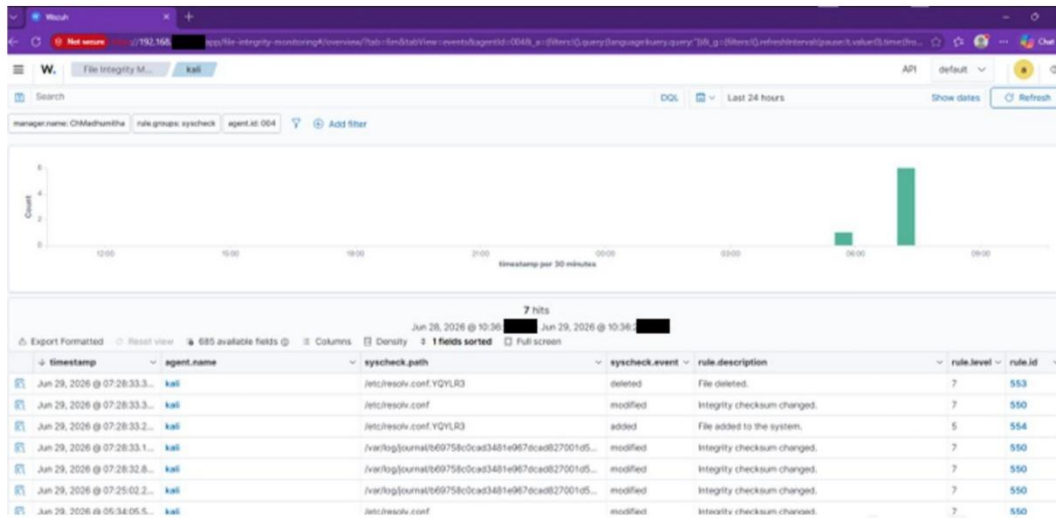


Figure 9 Wazuh Security Alerts

10. Results

The implementation was completed successfully, and all objectives were achieved.

The following results were observed:

- Wazuh Virtual Appliance was deployed successfully.
- Wazuh Manager was configured correctly.
- Windows endpoint was successfully connected.
- Linux endpoint was successfully connected.
- File Integrity Monitoring was enabled.
- Test file modifications generated security events.
- Alerts were successfully detected and displayed in the Wazuh Dashboard.
- Centralized monitoring of multiple endpoints was successfully demonstrated.

Overall, the project confirmed that Wazuh can effectively monitor endpoint activities and provide real-time visibility into security-related events.

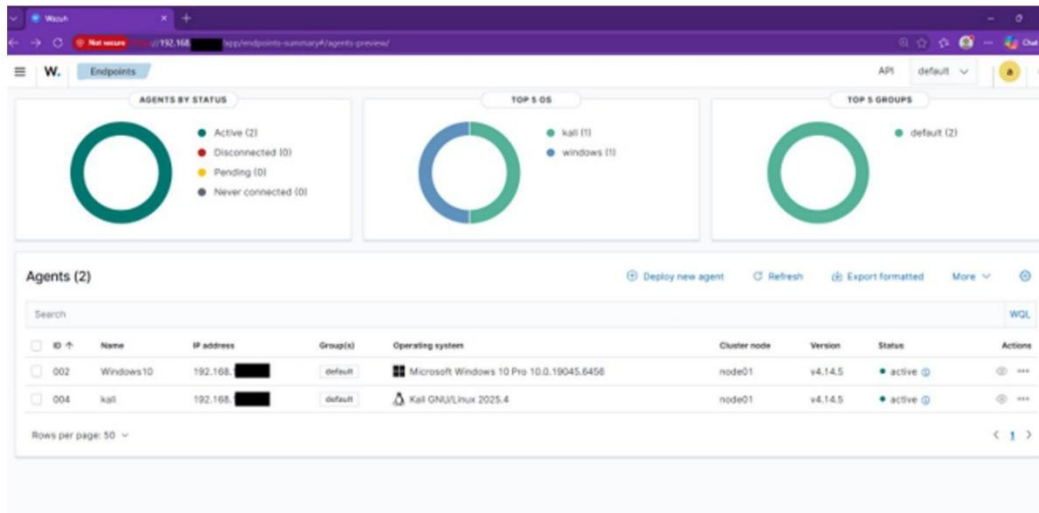


Figure 10 Final Dashboard Showing Active Agents and Alerts

11. Challenges Faced

During the deployment process, several common configuration challenges were encountered.

Some initial issues included ensuring proper communication between the Wazuh Manager and Agents, verifying the correct Manager IP address, restarting agent services after configuration changes, and confirming that monitored directories were correctly configured for File Integrity Monitoring.

Each issue was resolved through systematic troubleshooting by reviewing configuration files, checking service status, validating network connectivity, and confirming successful agent registration in the Wazuh Dashboard.

These troubleshooting steps improved my understanding of SIEM deployment and endpoint management.

12. Learning Outcomes

By completing this project, I gained practical knowledge in the following areas:

- Fundamentals of Security Information and Event Management (SIEM)
- Architecture and components of the Wazuh platform
- Deployment of the Wazuh Virtual Appliance
- Configuration of Wazuh Manager and Wazuh Agents
- Windows and Linux endpoint monitoring
- File Integrity Monitoring (FIM) concepts

- Security event generation and validation
 - Alert analysis using the Wazuh Dashboard
 - Centralized log monitoring
 - Basic Security Operations Center (SOC) workflow
 - Hands-on experience in Blue Team monitoring and detection techniques
-

13. Conclusion

This project successfully demonstrated the deployment and configuration of a centralized SIEM solution using Wazuh. The implementation included setting up the Wazuh Virtual Appliance, connecting Windows and Linux endpoints, configuring File Integrity Monitoring, generating security events, and validating alerts through the Wazuh Dashboard.

The practical experience provided valuable insight into how Security Operations Centers monitor endpoint activity, detect suspicious behavior, and investigate security events. Working with Wazuh strengthened my understanding of endpoint security, centralized log management, security monitoring, and Blue Team operations. This project serves as a strong foundation for future learning in threat detection, incident response, and advanced SOC analysis.
